

limits. The actual observed bandwidth depends on the number of devices in a system and additional bus traffic caused from any overhead. If a lot of data needs to be streamed on a large number of channels, bandwidth may be the most important consideration while choosing the data acquisition bus.

What are my single-point I/O requirements?

Applications that require single-point reads and writes are often dependent on I/O values to be updated immediately and consistently. Based on how the bus architectures are implemented in both hardware and software, single-point I/O requirements could be the determining factor for the bus that you choose.

Bus latency is the responsiveness of I/O. It is the time delay between calling a driver software function and updating the actual hardware value of the I/O. Depending on the bus you choose, this delay could range from less than a microsecond to a few milliseconds. In a proportional integral derivative (PID) control system, for example, this bus latency can directly impact the maximum speed of the control loop.

Fig. 1 shows a common block diagram for a feedback control system where the compensator (or controller) sends an output signal to the system or plant, and reads back a single-point sensor value to calculate the error in the process. If there are many delays along the communication bus, the time between output updates and sensor measurements also increases, resulting in a higher amount of error in the control system.

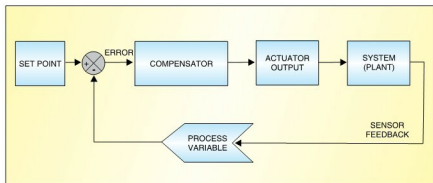


Fig. 1: Basic feedback control system where the controller relies on consistent single-point sensor measurements to correctly control the system or plant

Common communication buses used for measurement and automation

- PCI
- PCI Express
- USB
- Serial
- PXI
- PXI Express
- Ethernet
- Wireless

Another important factor in single-point I/O applications is determinism, which is a measure of how consistently I/O can execute on time. Buses that always have the same latency while communicating with I/O are more deterministic than buses that can vary their responsiveness. Determinism is important for control applications because it directly impacts the reliability of the control loop, and many control algorithms are designed with the expectation that the control loop always executes at a constant rate. Any deviation from the expected rate makes the overall control system less effective and less reliable. Therefore buses that

are high in latency with poor determinism—such as serial, Ethernet, or USB—should be avoided when implementing closed-loop control applications.

The software side of how a communication bus is implemented plays a big part in bus latency and determinism. Buses and software drivers that have support for real-time operating systems provide the best determinism and therefore give you the highest performance. In general, internal buses such as PCI Express and PXI Express are better for low-latency single-point I/O applications than external buses.

Do I need to synchronise multiple devices?

Many measurement systems have complex synchronisation needs, whether it is synchronising hundreds of input channels or multiple types of instruments. A stimulus-response system, for example, might require the output channels to share the same sample clocks and start triggers as the input channels to correlate the I/O, and better analyse the results. Data acquisition devices on different buses provide different ways of accomplishing this. Almost

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